THE G-RING CONDITION CAN BE CHECKED AT MAXIMAL IDEALS ONLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The G-ring condition, meaning that the formal fibres of the local rings are geometrically regular. defined in the same section, source lines 12179-12261. Supplies the property that has to be verified at every prime.
- Complete Noetherian local rings have regular formal fibres. the author cites Proposition proposition-Noetherian-complete-G-ring at proof lines 12409-12411. Makes the completion of the completion regular.
- Regularity of ring maps composes, and regularity of a map can be tested after factoring through an intermediate ring. the author cites Lemma lemma-regular-composition and Lemma lemma-regular-permanence at proof lines 12413-12425. Carries the regularity from the corner rings back to the local map at the chosen prime.

Source reference. Original lemma 07PT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9808–9808. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9807–9807. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9833–9833. Navigation pointer to a statement the

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Source reference. Original lemma 07NT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9773–9773. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Then R is a G-ring if and only if $R_{\mathfrak{m}}$ has geometrically regular formal fibres for every maximal ideal $\mathfrak{m}$ of R.*

Proof. Assume $R_{\mathfrak{m}} \to R^\wedge_{\mathfrak{m}}$ is regular for every maximal ideal $\mathfrak{m}$ of R. Let $\mathfrak{p}$ be a prime of R and choose a maximal ideal $\mathfrak{p} \subset \mathfrak{m}$. Since $R_{\mathfrak{m}} \to R^\wedge_{\mathfrak{m}}$ is faithfully flat we can choose a prime $\mathfrak{p}'$ in $R^\wedge_{\mathfrak{m}}$ lying over $\mathfrak{p}R_{\mathfrak{m}}$. Consider the commutative diagram

$$\begin{array}{ccccc} R^\wedge_{\mathfrak{m}} & \longrightarrow & (R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'} & \longrightarrow & (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'} \\ \uparrow & & \uparrow & & \uparrow \\ R_{\mathfrak{m}} & \longrightarrow & R_{\mathfrak{p}} & \longrightarrow & R^\wedge_{\mathfrak{p}} \end{array}$$

By assumption the ring map $R_{\mathfrak{m}} \to R^\wedge_{\mathfrak{m}}$ is regular. By Proposition 07PS $(R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ is regular. The localization $R^\wedge_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'}$ is regular. Hence $R_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ is regular by Lemma 07QI. Since it factors through the localization $R_{\mathfrak{p}}$, also the ring map $R_{\mathfrak{p}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ is regular. Thus we may apply Lemma 07NT to see that $R_{\mathfrak{p}} \to R^\wedge_{\mathfrak{p}}$ is regular. □